Real-time Double Reed Shaper 3D Modeling

CurvaSURE Shaper Generator [v4]

User Manual

Welcome to CurvaSURE! This application empowers you to design and generate precise 3D printable straight shaper forms for various double reed instruments as well as exporting files that can be sent away to make metal versions. CurvaSURE offers an enormous repository of well-established professional shapes or allows you to create your own!

1. Getting Started

Upon launching the application, you'll see a main window divided into two sections:

- **Left Panel (Control Panel):** Contains all the input fields, parameters, and action buttons. This is where you define your shaper.
- **Right Panel (Preview Area):** Displays a real-time 2D visual representation of your shaper as you adjust parameters.
- ⚙️ **(Gear)**: Settings found in the upper left corner of the **Control Panel**. You can access the manual, show the license agreement and contact support.
- **v4.0.0 :** The version number found in the upper right corner of the **Control Panel.** This informs you as to which version of CurvaSURE you are running.
    - **Auto-update:** CurvaSURE now has an auto update system which will alert you to a new version, if available, the next time you open the program. It will also include release notes for what has changed.

2. Main Sections

2.1. Shaper Parameters

This section allows you to define the fundamental dimensions of your shaper.

- **Preset:**
    - **Select a Preset:** This dropdown menu provides a comprehensive list of pre-defined shaper profiles categorised by instrument (e.g., Bassoon, Historical

Bassoon, Contrabassoon, Contraforte). Selecting a preset will automatically populate all relevant parameters and shape points, giving you a great starting point.

- ***Online Presets:*** The application automatically checks for and downloads new or updated presets from an online repository each time you open the program. You can also manually trigger a check using the “Download/ Update Presets (Online)“ button in the Preset menu.

Preset names referencing well-known shaper profiles (e.g., Rieger 1a, etc.) are used solely to describe the style or geometry of the profile for user clarity. All measurements are either derived from publicly available information or independently taken.

These references are for identification purposes only and do not imply any affiliation with, endorsement by, or licence from the original manufacturers.

- **Open Custom Preset:** Use this to load a previously saved **.json** preset file.
- **Save Custom Preset:** Once you've designed a custom shaper you're happy with, use this feature to save all its parameters and shape points to a **.json file**. This allows you to easily recall your custom design later or share it with others. ***Important:*** *No custom presets are stored in the program memory, only locally on your hard drive.*
- **Download/Update Presets (Online):** Selecting this checks the online repository for any un-downloaded presets or updates to the presets that may have been published. The program will automatically check every time it opens, this is simply a way to manually engage that check.
- **Load Backup Database (Offline):** In the event that the program is unable to access the online repository, an offline backup can be loaded through this interface. Please get in contact if this is needed.
- **Clear Preset Cache:** This removes all the plugins cached on your computer. If you are having an issue with a preset or if you know a preset has been updated but doesn’t seem to be showing, click ‘Clear Preset Cache’ and then ‘Check for Updated Presets Now’. This will download the newest versions of the presets into the program. (*This will not affect the presets you have exported and saved elsewhere.)*

- **Preset Search (🔍):** Clicking this will open the search function. This allows you to search the entire repository of shapes very easily. You can also filter by measurements of flat tip or clipped flat tip.
  - ***Flat Tip:*** If you wanted a bassoon shape in the range of a 15.5 - 15.6mm flat tip, enter the minimum and maximum measurements with those values and all the shapes that fit within that range will appear.
  - ***Clipped Tip:*** If you normally clip your reed at 3mm and want a tip measurement of 15.5 to 15.7, you can enter all three values and see what appears. This can be useful to find similar shapes to experiment with.

- **Total Cane Length (mm):** This defines the overall length of the cane for which the shaper is designed. There are standards (such as bassoon being 120mm, contra being 150mm, etc) but ***any value can be entered in the text field between 50 and 180mm.***

- **Gouge Diameter (mm):** Specifies the diameter of the gouge to embed in the shaper. To be as precise as possible and to prevent your cane from being unnecessarily bent, it is generally recommended to select the correct gouge you use.

    - ***Gouge Compensation:*** The program will automatically calculate the shaper width measurements needed to give you the correct shape based on the flat width. This means that if you change the gouge diameter, the shaper width numbers will change due to the cane wrapping around the shaper at slightly different dimensions.

    - ***Note:*** *for gouges of* ***16mm****,* ***18mm****, and* ***20mm****, the concave shaper half uses a slightly smaller gouge so that it grabs the cane better. This is expected. The convex half is still the set value.*

- **Bolt Hole Diameter (mm):** Sets the diameter for the bolt holes on the shaper which clamp the two halves together. The default value is **M5 Hex (5mm)** bolts, but there are more options for both metric and imperial measurements. There is an inbuilt **0.2mm** addition to this diameter to make threaded bolts slide through easier and to compensate for printing layers. This would make a 4mm hole actually 4.2mm.

    - **Hex Bolts:** In general, hex bolts are recommended as there will be an indent created in the bottom of the shaper to nest it. This helps with tightening and doesn't require additional adhesion.

    - **Spring Indent:** As a default, there is a slightly larger sunken area around the hole on the inside of the shaper halves. This is for spring placement, if you would like to make the shaper halves easier to use. **0.6 x 7 x 12mm** compression springs are what this indent is configured for. If springs are not used, these indents will not affect functionality.

- **Generate Guide Indents:** These checkboxes control whether small physical indents are included at the tip **(Tip)**, narrowest point **(NP)**, and butt **(Butt)** of the shaper. The first checkbox controls whether any indents show up. These indents serve as visual and tactile guides for accurate cane placement and shaping.

- **Guide Indent Placement:** This checkbox controls whether the position of the Guide Indents (should you have this enabled). You have the option of placing the indents along the edge of the cutting surface (**Side**) or on the Top/Bottom of the shaper (**Top**).

- **Generate Alignment Posts (Recommended):** This checkbox controls whether alignment posts are embedded in the shaper to assist with cane placement. It is recommended this stay ticked as it will assist in both shaper alignment and strength. This is recommended to stay on, unless perhaps you print vertically.

- **Engrave Shaper Name (Recommended):** This checkbox controls whether the name of the preset is engraved on the bottom of the shaper. It is recommended to keep this enabled so as to easily identify the shaper after printing.

- **Shaper Name (Editable):** This box allows for customisation of the engraved shaper name, assuming *Engrave Shaper Name* is ticked. The box will auto-populate with the

name of the loaded preset but it can be changed to whatever is desired. It is recommended to keep the character count, including spaces, to around 12 to maintain readability.

2.2. Shape Points (Customisation Hub)

This is where the true flexibility and power of CurvaSURE shines, allowing you to define highly custom shaper profiles. The shaper's profile is determined by a series of (Distance, Shaper Width and Flat Width) points.

- **Folded Tip Width:** This shows the actual width of the reed tip when folded. When you form a blank, the gouge collapses into a flat line so the actual tip width of the blank will always be wider than a shaper that has the gouge built into it. This is helpful to better visualise the resulting tip opening.
- **"0 (Tip)":** This is the starting point of your shaper profile, representing the width at the very tip of the cane.
- **Intermediate Points:** These are user-defined points along the length of the shaper.
    - **Distance (mm):** This value specifies how far along the shaper's length (from the tip) the point is located.
        - **Accepted Range:** Values must be between **0mm and 90mm**.
    - **Shaper Width (mm):** This value defines the width of the cane at the specified distance from the tip of the reed.
        - **Accepted Range:** Values must be greater than **0mm and up to 30mm**.
    - **Flat Width (mm):** This value defines the width of the cane if the shape were flattened. If you have measurements from a fold-over shaper, these can be entered directly in this field and the corresponding straight shaper values will automatically be calculated.
    - **Add Point:** Click this button to add a new intermediate point row. You'll enter a "Distance from Tip (mm)" and a "Width (mm)". To insert a row at a specific point, click in the field above where you want to insert a point, then click 'Add Point'.
    - **Remove Point:** To remove an intermediate point, first **click on either the "Distance from Tip" or "Width" field** of the specific intermediate point you wish to remove. Once selected (the field will have focus), click the "Remove Point" button. You cannot remove the "Tip" or "Butt" required points.
    - **Undo/Redo:** These buttons trigger the Undo/Redo function for the shape points. Standard keyboard shortcuts may also be used (**Mac:** *Cmd+Z, Cmd+Y*, **Windows:** *Ctrl+Z, Ctrl+Y*)
- **"— (Butt)":** This is the final point of your shaper profile, representing the width at the butt end of the cane. If you have a blank template selection, its "Distance from Tip" is automatically calculated as half of the "Total Length (mm)". If you have selected a preset and not a template, the '(Butt)' text will not appear.

Input Validation: The application performs real-time validation of your input values. If a value is outside the accepted range or in an invalid format, the input field will turn red, and an error message will appear in the preview area or via a pop-up.

2.3. Curve Options

This section controls how the application interpolates between your defined shape points.

- **Smooth Curve (Recommended):** This option uses a cubic spline interpolation method to create a smooth, continuous curve between your defined points. This is generally recommended for most shaper designs as it produces a more organic and flowing profile. To get the benefit of the curve, multiple points are needed. Generally speaking, the more points you have the better your shape will turn out.
    - As you're inputting points, you may notice that the Smooth Curve option creates some bizarre shapes. This is only the mathematics trying to make sense of the overall shape of the points currently available. As you work your way down the shaper, the wildness of the curves will be tamed - you simply need more data points.
    - ***Tip:*** When creating your own shape from scratch, choose the **Straight Lines** option. This will help you sketch the general shape out. Once you have enough points, switch to the **Smooth Curve** option and edit from there.
- **Straight Lines:** This option connects your defined points with straight line segments. This can be useful for creating more angular or segmented shaper profiles, particularly with fewer Shape Points. This is also useful for experimenting with 'rough' or 'experimental' shapes prior to engaging the smooth curve, as you can input fewer points and still have a workable result.

If you have enough points listed, the differences in the Smooth Curve and Straight Line settings become very small to negligible.

2.4. Advanced Features

- **Offset Shape Start:** This slider allows the start of the shape to be pushed back or pushed up a specified amount from the tip, with retaining the proportions of the original shape.
    - ***Example:*** If you wanted to retain the tip width of the shape after clipping 2mm off the tip, you would offset the shape start by 2mm.
    - ***Example:*** If you are using a shape that was a foldover originally, you might want to set the offset to the value your cane sticks up above the tip of the foldover shaper. This is usually around 0.5-0.7mm. That way you will have a more accurate shaping result to the original shape.
    - This setting *will* be saved if you export a preset.

- **Global Width Change:** This slider allows you to change the overall width of the shape by the amount set on the slider. You might find yourself wanting a slightly wider or narrower shape overall - this dialogue allows for you to experiment without having to go through and manually adjust the shaper values.
    - When you engage the slider with a positive or negative value, options will appear below to apply the width change to either the Shaper Width or the Flat Width. **The change will apply to whichever option is selected** - you do not need to click the Apply button below unless you want to permanently change the points.
    - The button **Apply Width Change to Shape Points** should only be clicked if you want the width change to be reflected in the actual shape points.
    - ***Note:*** When exporting a shape as an STL, DXF or saving a preset - **the width change will be maintained** even if you've ***not*** pressed the apply button. This allows for saving a shape and experimenting while keeping the integrity of the original measurements.
- **Enable Asymmetrical Shape:** This tick box allows the user to adjust the left and right side of the shape independently. Once you tick the box, you will see that the Left Profile and Right Profile tabs become active and two more buttons appear in the Shape Points area: **Load Preset as Side** and **Mirror to Opposite Side**.
    - **Load Preset as Side:** When you toggle this, you may go up to the preset menu and select a preset (internal or loaded custom). The values of this preset will then be loaded into whichever profile tab you have selected (Left or Right). This function will remain active until you select a preset or you click the button again, which should currently display **Cancel Load as Side**.
    - **Mirror to Opposite Side:** When you click this button, the tab that is currently displayed will copy the values it contains to the opposite side. For example, if you have the Left Profile selected, the values would then be copied into the Right Profile.
    - When you **disable** asymmetry, the program uses the Left Profile as the points to construct the symmetrical shape.
- **Display Half Width Values:** This tick box becomes available to use when Enable Asymmetrical Shape is ticked. This changes the values in the Shape Points dialogue to half their widths. This is helpful in creating asymmetric shapes as the half widths will be the distance from the midpoint line to the outside edge.
- **Generate Shaper Ears:** Ticking this box will make ears appear around the tip. On many oboe and cor anglais fold-over shapers, this is a popular feature, providing extra cane around the tip when shaping. The default width is 0.5mm and length is 7mm, but these values can be changed (**maximum 0.5mm width** and **5-10mm length range**)
- **Scale Shape:** This tick box will display the scaling dialogue. This feature allows the user to select any shape and scale it up or down in any way they might like.
    - **How to Scale:** Select your desired shape (for example, a bassoon shape) and select **Scale Shape**. As bassoon shapes are usually 120mm, you should see 120mm as the total length value. If you then want to scale this shape to contrabassoon size (usually 150mm), you would then enter 150mm in the total

length field and hit apply.

You will notice that the tip width in the scale dialogue changes value when you change the length - reflecting the scaled up size. If you decide that the scaling caused the shape to be too narrow or wide, you can untick the **Lock ratio** box and the Width and Length scaling are able to be used independently.

You can use any of the four boxes to scale the shape (if the ratio is locked) but it might be helpful to start with the Length (mm total) box, as you can be certain it will scale to your desired length. All other boxes will adjust to whichever values are input. (It's much easier to know that a bassoon shape scaled to contrabassoon should be 150mm in length than it is to know the ratio is 1.25!)

- ***Note:*** Scaling will not occur unless you click **Apply Scale to Shape**.
- ***Note:*** The scaling ratio will always apply to the base shape that is loaded. If you scale a bassoon shape to a contrabassoon shape (1.25x) and you want to change it back, simply enter 1.0 in the ratio or change back to the original length. Hitting **Apply Scale to Shape** more than once won't cause compounding scaling.

2.5. Preview Controls

The right panel provides a visual preview of your shaper.

- **Display Preview using Flat Width:** Ticking this box will display the preview as if the shape was completely flattened. This is purely a visual change, helping the user to visualise what a completely flattened piece of cane would look like. *This does not affect the file generation.*
- **Show Guide Marks:** Toggles the display of helpful visual guides on the preview:
  - **Red Dashed Line:** Represents the centre line of the shaper.
  - **Green Dashed Line:** Mark the butt end of the shaper.
  - **Orange Dashed Line:** Indicate the narrowest point of the shape.
  - **Purple Dashed Line:** If Offset Shape Start is engaged, this indicates the place to where the tip measurement is offset.
- **Show Point Marks:** Ticking this will display small black dots along the shaper outline. This is primarily to help determine how the points are affecting the overall shape, particularly if the Smooth Curve option is enabled.
- **Show Midpoint Line:** Ticking this will display a dotted line down the centre of the shaper. This is particularly helpful for visualising asymmetrical shapes. This box will be ticked automatically when asymmetrical shapes are enabled.
- **Auto-update:** When enabled, the preview will automatically regenerate in real-time as you modify parameters or shape points.

- **Generate Preview:** If "Auto-update" is disabled, click this button to manually refresh the preview.

- **Show Clipped Tip at __ mm:** Ticking this box and entering a value will display a dotted line at the specified distance from tip. This line represents where you will clip the reed after forming.

    - An annotation box will appear labeled **Clipped Tip**. Inside this box will be the **distance** from the tip, the **shaper width** at the clipped tip, and the corresponding **flat width**.

- **Show Collar at __ mm:** Ticking this box and entering a value will display a collar line at a specified distance from the tip. This can help visualise your profile will look on the shaper.

    - As above, an annotation box will appear with distance, shaper width and flat width values.

    - Ticking both **Clipped Tip** and **Collar** will **calculate the total blade area** between the collar and the tip and appear in the **Clipped Tip** annotation box.

- **Show Curvature Comb:** The Curvature Comb is a diagnostic tool that helps you visualise the smoothness of your shaper's profile when using the "Smooth Curve" option. When activated, it displays a series of lines, or "teeth," perpendicular to the shaper's outline. The length of each tooth is directly proportional to how sharply the shaper is curving at that point. Longer teeth indicate a tighter curve, while shorter teeth show a flatter, gentler curve. A smooth and gradual transition in the length of the teeth signifies a well-formed curve without any unwanted bumps or kinks. A perfectly straight line should have no teeth visible.

    - You can adjust the comb using the controls in the "Preview Controls" section. The **Comb Density** slider changes the number of teeth shown, providing a more or less detailed view of the curve. The **Comb Scale** slider acts as a magnifying glass, increasing or decreasing the length of the teeth. Increasing the scale can help exaggerate and identify very subtle imperfections in the curve that might otherwise be hard to see, allowing you to fine-tune your points for a perfectly smooth shaper outline.

- **Load Preset as Reference:** Ticking this box will enable the Load Preset as Reference functionality, which allows for two shapes to be overlaid and compared with one another.

    - Tick the box to enable, click **Load Preset as Reference** and go to the menu, selecting any preset as you normally would. The preset will then be loaded as a reference shape behind the current shape. To switch between shapes, click between **A** and **B** either in the Preview Controls or in the upper right of the display area, beneath the pan controls.

    - The **A shape** is always the shape **you are working on** and the **B shape** is the **reference**. Whichever shape is selected will be shown as the normal **blue outline** in the front with the other shape as a dotted **orange outline** behind it.

- Ticking **Show Clipped Tip** and **Show Collar** will display different annotation boxes for each shape, allowing you to compare the widths and area for each shape.
- Hovering your mouse over the shape will **display the widths for whichever shape is currently in the foreground**.
- To disable the background shape, either untick the box (to hide the shape) or click **Clear Reference**, which removes that preset from the reference field.

- **Additional Preview Controls**
  - **Zoom / Pan *(upper right corner of the preview area)*:** These are buttons which allow you to increase the size of the preview as well as moving around. The **Pan** controls are arrows that allow you to move around the image. The central circular button will both center and zoom out the image. There are **Zoom** and **Pan** buttons which allow you to increase the size of the preview as well as moving around. The **Pan** controls are arrows that allow you to move around the image. The central circular button will both center and zoom out the image.

    Note: You can also zoom using your mouse scroll wheel and pan by grabbing the shaper/grid and moving it around.
  - **Width, Half Width, and Distance Measurements:** Additionally, if you **hover your cursor over the outline** you will be able to see the width, half width, and distance at any point along the shaper.

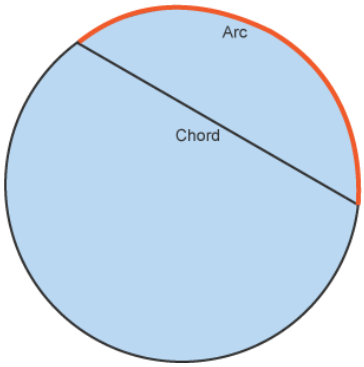
2.6. Action Buttons

- **Generate Selected Files:** This is the core function for exporting your shaper design.
  - Clicking this button will prompt you to choose a save location.
  - The application can generate two types of files. Tick either or both to generate:
    - **STL (.stl):** A 3D model file suitable for 3D printing. The generated STL will include two mirrored shaper halves and a connecting base, ready for printing.
    - **DXF (.dxf):** A 2D CAD file representing the outline of your shaper, useful for laser cutting or further CAD work.
  - **Error Handling:** The application will validate all parameters before generation. If any required parameters are missing or values are invalid, an error message will be displayed.
- **Help:** Links back to this very manual!

- **Shaper Measurement Converter *(Legacy Tool)*:**

  [As of CurvaSURE v3.0.0, this is no longer necessary due to the added Flat Width column.]

  This is a computational tool that will convert measurements between straight shapers and fold-over shapers or flat reed measurements. Straight shapers maintain the gouge radius while you shape the cane whereas fold-over shapers force the cane at the tip flat (or almost flat). This tool accounts for the differences in curvature of the cane, so if you wanted to input the measurements of a fold-over shaper, you would get the correct straight shaper values.



Example: Let's say you wanted to convert a Rieger 1A shaper tip to a straight shaper. On the Rieger website (and confirmed by measuring the actual shaper tip) the tip width is **15.5mm**. If you shape a piece of cane using this tip, the resultant tip will be 15.5mm as the fold-over shaper more or less resembles the folded state of the reed.

However, if you were to apply those measurements to a straight shaper with no adjustment, then the resulting measurement of the folded tip would be an incorrect width of **16.18mm** (using a 32mm gouge).

This is because the **chord length** (the width measurement of a piece of cane measured from above or the straight shaper width) was used instead of **arc length** (the total length of the flattened out tip).

We want a consistent tip measurement regardless of our gouge settings, therefore the **arc length** *must* stay the same and the gouge will be the same. This only leaves the **chord length** variable, so to achieve an accurate shape transfer it is this value that needs to be adjusted.

Therefore, if you have a shaper tip you would like to convert into a straight shaper, take measurement values at the tip and every 5-10mm, plug those values into the fields provided along with the desired straight shaper gouge, and press **Calculate**. Add these values to the Shape Points field and voilà, you have a straight shaper copy!

- If there is a slight curvature at the back of the shaper tip, you can choose to take that as a flat-measurement and accept the slight resulting width difference (often less than 0.1mm) or you can take a radius gauge measurement at various increments and plug different values in. For example, if the back of a shaper tip has a very slight bend (i.e. 40mm radius), then plug in 40mm as the gouge and take the overhead measurement and enter that value into the Straight Shaper field. Convert and get the Flat Measurement. Next, erase the Straight Shaper value and change the Gouge value to match your desired shaper (while keeping the Flat Measurement you had) then generate the new Straight Shaper value. It's a roundabout process to do it this way, but you're guaranteed to convert your shaper correctly!

- **Order at Reeds n' Stuff:** Clicking this button will direct you to the Reeds n' Stuff website, displaying the CurvaSURE order page. On this page you will be able to order 3D printed shapers, if you don't currently own a printer (exporting the **STL** file). You will also be able to order metal versions of the shapers you designed (exporting the **DXF** file).
    - ***Note:*** Before ordering, please check what gouge settings, length, and bolt holes you are using. In order for Reeds n' Stuff to accurately print or machine your shapers, you must use the settings they mention.

3. Tips for Best Results

- **Validate Inputs:** Always ensure your numerical inputs are within the accepted ranges. Invalid inputs can lead to unexpected or unprintable shapes.
- **Start with 'Straight Lines' Enabled:** Due to the nature of how the spline curve works, the first values entered can make a strange visual. If you enable **Straight Lines** first and then sketch out your general shape, you'll have a lot of success when engaging the **Smooth Curve** option.
- **Preview Regularly:** Even with auto-update, occasionally review the preview to ensure the shape is developing as intended.
- **Understand Your Cane:** The ideal shaper profile depends heavily on the specific cane you are using (its thickness, density, and natural curve). Experimentation and iterative design are key.
- **3D Printing Considerations:** When generating STL files, remember that the final print quality will depend on your 3D printer's capabilities and slicer settings. Ensure you use appropriate materials and print resolutions.

4. Troubleshooting

- **"Input Error" / "Plotting Error":** Check the values you've entered in the "Shaper Parameters" and "Shape Points" sections. Look for red-highlighted fields.
- **"Cannot Remove Point":** Remember you can only remove intermediate points, not the "Tip" or "Butt" required points. Ensure you've clicked on an intermediate point's field to select it before attempting to remove.
- **"Failed to generate files":** This usually indicates missing or invalid parameters. Review all fields, especially those related to "Total Length", "Gouge Diameter", and "Screw Hole" dimensions.
- **No Logo/Icon:** If the logo or icon doesn't appear, it simply means the image files are not found in the expected location. The application will still function normally.

- **"Connection Error" / "Data Error" for Presets:** This indicates an issue connecting to the online preset server or a problem with the downloaded data. Check your internet connection.

- **“Duplicate distance values are not allowed”:** This means that in your shape points, there are two width values listed at the same distance. The program doesn’t know which value takes priority, so one will need to be removed. ***Note:*** if you have changed the total cane length of a shape, you might unintentionally get a duplicate value if the new butt distance (which changes automatically to match the length) overlaps with a previously existing value. The same rules apply here - one value will need to be changed, but the selected value will need to go in the (Butt - Required) field.

5. Materials, 3D Printing Tips, Assembly and Shaping

5.1. What You Will Need

Below you will find the materials needed for fully assembling the straight shaper.

- **3D Printer:** Any sort of 3D printer will do, as long as the printing bed is large enough to be able to print the Total Length parameter. The Bambu Lab A1 or A1 Mini is a great choice to start off with, if it happens to be in your price range.

- **Filament:** In testing 3D printed shapers out, the best performing filaments of the group were **PLA**, **PLA+**, and **PETG**.

    - **PLA/PLA+:** PLA is a very commonly used filament praised for the high tensile strength it provides and evenness of printing surface. PLA+ is a modified version of this meant to be slightly stronger and more flexible. Printing with PLA/PLA+ is a great choice to start off with, but be aware that it can flake if you dig a knife into it. Care needs to be taken when shaping to prevent digging into the filament surface. Sandpaper is a useful tool to smooth areas out you can’t reach with a knife.

    - **PETG:** This is a filament with much higher flexibility than PLA/PLA+ but it is much more resistant to knife scraping. Due to the strength it possesses, it is often used for manufacturing parts that go through wear and tear. Depending on the infill percentage you end up using, you may find that if you can twist it ever so slightly but this is not an issue when the other half is attached and tightened. ***Warning:*** this filament is naturally much ‘stickier’ than PLA/PLA+ so you may have to clean the hot end of your printer more often or possibly purchase a hot end with a different tip.

  Any of these are good options, so it is worth experimenting to find out which you prefer.

- **Bolt and nut (at least 40mm in length)**: For clamping the two halves together, a bolt and nut are needed. The appropriate length for this is a minimum of 40mm as it will reach though the shaper giving just enough threading on the other side for the nut to close around. The bolt size needs to match the Bolt Hole Diameter option that you chose when generating the shaper.

- ***Recommended:*** Print a Hex Nut Grip which can house the nut meant to screw the bolt in. This is much easier to tighten and handle than using the nut on its own. Please click **here** to download the recommended STL files. After printing two of them, insert the nut into the bottom of the print and screw into the bolt. *Note:* if you have calipers, then it's a good idea to measure the flats of the bolt against the inside of the Hex Nut Grip to see if it needs to be scaled. A tighter fit is better!

- **Compression Spring 0.6 x 7 (or 8) x 15mm (optional)**: For additional ease of keeping the shaper open, a compression spring may be used. The shaper has embedded indents around the bolt hole where these springs may be placed. The diameter of the indent is 8mm, therefore any spring with that diameter will fit. However, the recommended height of the spring is 15mm. This will allow the spring to compress fully when closed without too much effort and have enough force to push the two halves apart when the bolts are loosened. Any spring thicker than 0.6mm will also be difficult to push closed.

- **Steel Shim Tape [30 microns or similar] (optional):** Using steel shim tape prevents your knife from cutting into the shaper filament at all, effectively transforming it into a metal shaper - at least in functionality. See section **5.4** for more information.

- **Digital Calipers (optional):** To easily measure reeds and also test the output of your prints, having digital calipers is highly recommended.

5.2. 3D Printing Tips

As straight shapers are fairly complex geometrical shapes, the question may arise how to get the best result printing them. Luckily there are only two options and both of them have their own merits. Experimentation to find out which works best for you is the way to go, but pros and cons of both options will be listed below based on experience.

- **Print Settings**: The following print settings are recommended for the best result.

  - ***15% infill, Gyroid or Grid*** - Gyroid is suggested as it won't cross over itself while printing.

  - ***Wall thickness*** - *3* layers at 0.4mm (1.2mm thickness in total)

  - ***Avoid crossing walls*** - If your printer has this function, then it is highly recommended that you engage it so as to prevent and filament build-up and inconsistencies in the print if the nozzle catches on a wall. This is especially important if you print vertically, as the shapers can easily topple.

  - ***Z seam*** - If your slicer has this option, select **Back** as the placement for the Z seam. This moves the position of where the layer printing starts to the end of the shaper. Many printers will start in the middle of the shaper around the tip, which can lead to inaccurate results.

  - ***Layer Height*** - As fine as your printer can do. Smaller layers will generally yield better accuracy.

- **Printing Horizontally (Default / Preferred)**: When you export files from CurvaSURE, the default printing layout is horizontal. This means the longest part of the shaper is printed on the bed.
    - This is an easy and consistent way of reliably printing. There is inherent stability in that the base is long and the print isn't too high.
    - One draw back is that the curvature of the inside will be slightly 'bumpy'. If you hold up the concave half of the shaper you might find that you can see small 'steps' as the curve goes up and down. This is simply due to the way 3D printers layer the filament. This can be lessened by creating thinner layers but it will always exist to a certain extent when printing in this orientation.
    - **Tip:** To counteract this 'step' appearance, a bit of sandpaper along the edges of will help, both with the halves open and closed together. If it has any effect on the shape, it is negligible - however, it does make shaping easier.

5.3. Assembly

Once you have all your materials and printed halves, it is time to assemble the shaper.

- **Step 1**: Determine which way you would like to orient the shaper. The shaper can 'smile' or 'frown', referencing the way the gouge is pointing based on the end of the shaper. As there are no threads embedded into the shaper bolt holes, you may choose which direction you prefer - however, it is recommended the shaper 'smile' (concave side on the bottom and convex side on the top) as it is much easier to rest the cane in when positioning. If you choose this orientation, the engraved shaper name should be facing up.
- **Step 2**: Take your bolts and insert them through the bolt holes from the bottom up through the top. The head of the bolt should be on the bottom of the shaper and the threaded ends should be sticking up through the top.
    - If you selected Hex Bolts in the program, there will be a hexagonal indent printed on the bottom of the shaper which allows for the bolt to fit within the shaper and not need additional fastening when opening or closing.
    - If you do *not* have a Hex Bolt and have a rounded top, then dab a bit of super glue around the top of the bolt, underneath the head, and then press and hold into the bottom shaper half until fastened. This is useful if you don't want the bolt to spin around while you are tightening the shaper.
    - If you choose not to glue it, when you tighten the nut on top place your thumb on the bolt head to prevent it from turning.
- **Step 3 (optional)**: If using springs, take the top half of the shaper off and drop a spring around each one of the threaded bolts sticking upwards. The springs should fit neatly into the indents on the inside of the shaper. Place the top half back on, being careful to line up the springs with the indents on that half.
- **Step 4**: Tighten a nut around each one of the upwards facing bolt threads, making sure that the shapers can close all the way.

- **Step 5**: If printing horizontally, it is recommended to rub sandpaper along the point where the two shaper halves meet when closed. This will smooth out the connection between the two halves and ensure nothing is sticking up to impede the shaping process.
    - Any change to the shape will be negligible, as you are only removing excess material that might be sticking up from the printing process. Be gentle and not heavy-handed.

5.4. Shaping

Due to the nature of 3D printing filament, there may be some concern over what using an abrasive, such as a knife or sandpaper, might do to the final shape. Based on testing, there hasn't been change to the overall dimensions of the shaper even after repeated uses.

This comes down to a few things:

- Using PLA+/PETG filament which seems to be more abrasive-resistant
- A shallow angle when cutting along the shaper (5-10 degrees above the shape)
- Using high grit sandpaper to gently smooth the print prior to cutting

If you take care while shaping to keep that small angle, then you will find it won't catch on the shaper.

Fail-safe solution: There is a method that will allow you to shape without worrying about cutting into the shaper at all. This method utilises **steel shim tape**. If you are able to source around 30 micron (0.03mm thick) shim tape (8mm wide), it can be placed on either side of the shaper cutting area to protect the filament underneath.

This only adds a total of 0.06mm to the shaper width which is a functionally negligible addition, so the shape should not need to be altered.

Tip: If you do want to be precise, you can remove the total combined thickness of the tape from the shape using the Global Width Change. In the example above, you would move the slider to -0.06mm and make sure it is being applied to the **shaper width**, as this is where the thickness from the tape is being offset.

A recommended brand is **Matrix Engineering *Super Shim Plus (0.03mm).*** This is a British brand and may not be available near you, so any comparable thickness / brand should work. If you are US-based, a similar brand is **Cutting Edge Grinding & Supply Co *Super Shim (.001")***.

6. Support

Any support inquiries or suggestions for features can be address directly via this form.

If the form isn't working or if you haven't heard back, please reach out via this website.

Thank you and enjoy CurvaSURE!